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Cognitively Assistive Robots at Home: Translating Clinical Interventions to Robots

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Abstract

Background: Much research in healthcare robotics explores extending rehabilitative interventions to the home. However, for adults, little guidance exists on how to translate human-delivered, clinic-based interventions into robot-delivered, home-based ones to support longitudinal interaction. This is particularly problematic for neurorehabilitation, where adults with cognitive impairments require unique styles of interaction to avoid frustration or overstimulation.

Method: For the past several years, our team has worked with neuropsychologists to develop cognitively assistive robots (CARs) that administer compensatory cognitive training (CCT) autonomously and longitudinally to people with mild cognitive impairment (PwMCI) at home. Our system helps users practice cognitive strategies to strengthen skills such as attention, learning/memory, and executive functioning.

We explore the design of robot-delivered neurorehabilitation interventions for PwMCI. We used these prototypes as design probes to understand how neuropsychologists (n=6) envision translating CCT to a CAR, and features the robot intervention needs to be successful, such as supporting goal setting, content personalization, encouragement for real-world transfer, and ways to longitudinally maintain engagement. We also conducted interviews with PwMCI (n=3), the end users of the robot-administered intervention.

Result: We report our findings and specify design patterns and considerations for translating neurorehabilitation interventions to robots. First, we provide insights for translating neurorehabilitation interventions to CARs in order to contextualize them to the lives of PwMCI. Second, we present new interaction design patterns for robot-delivered neurorehabilitation interventions to maintain longitudinal engagement and intervention efficacy. Finally, we propose design considerations for developing robots for PwMCI, a population with unique needs and abilities distinct from those of people with dementia and older adults [1].

Conclusion: This work will serve as a basis for future endeavors to translate cognitive training and other clinical interventions onto a robot, support longitudinal engagement

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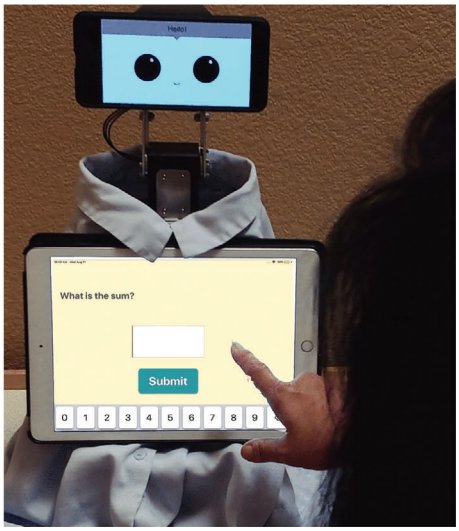
with home-deployed robots, and ultimately extend the accessibility of longitudinal health interventions for people with cognitive impairments.

[1] Kubota, A., Cruz-Sandoval, D., Kim, S., Twamley, E., and Riek, L.D. "Cognitively Assistive Robots at Home: HRI Design Patterns for Translational Science." In *Proceedings of the 17th Annual ACM/IEEE Conference on Human Robot Interaction (HRI)*. 2022.



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